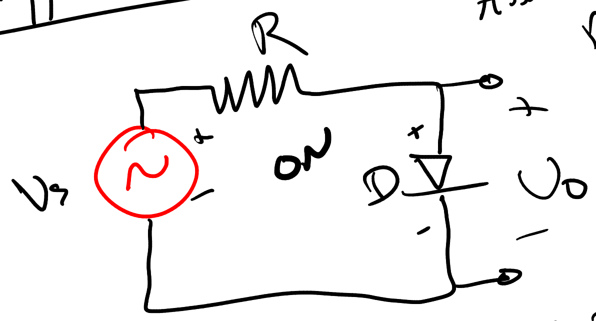
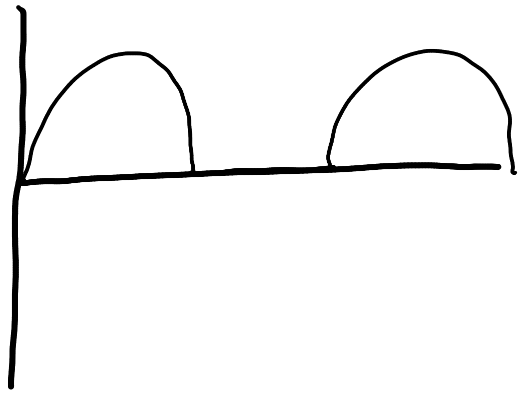
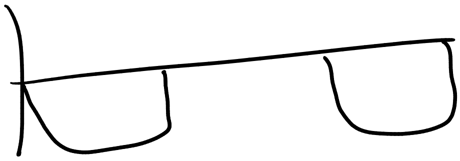
Clipper Circuits

Assume Ideal Diodes

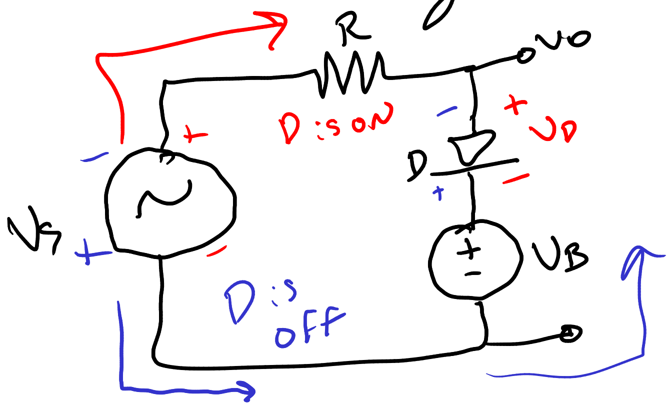


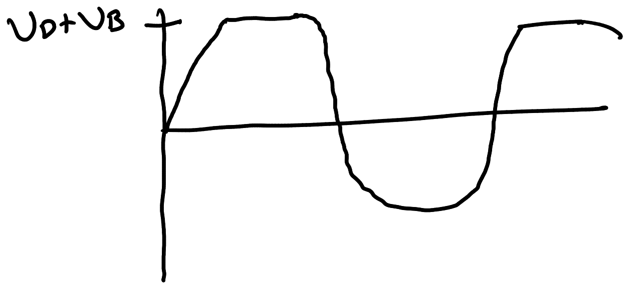


* What happens if we swap diode?

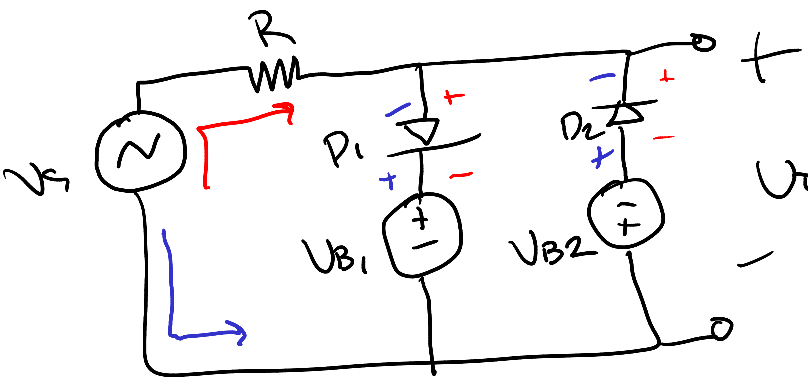


* Now add a battery and let's assume CVD 7





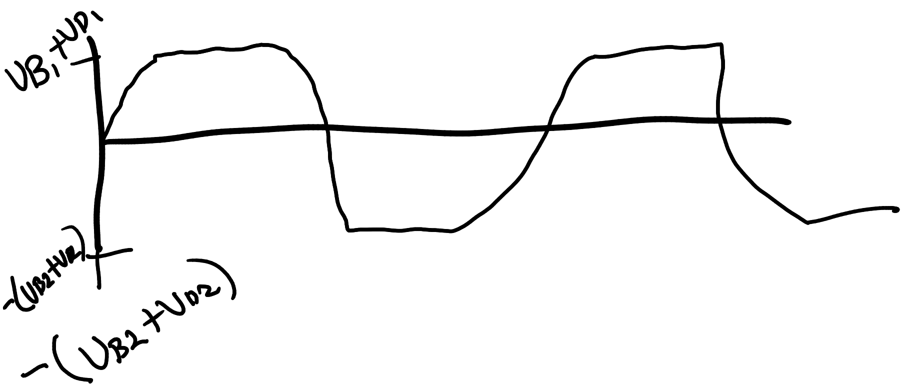
* D is off as long as $V_s < V_B + V_D$, output follows input
* When $V_s > V_B + V_D$, D is on and output is clipped to $V_B + V_D$



Parallel Clipper

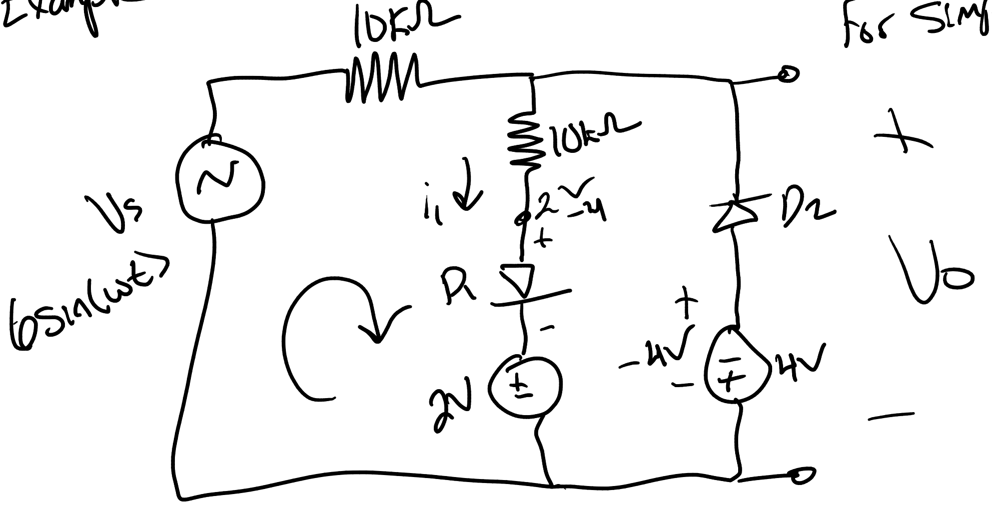
D_1 is ON D_2 OFF

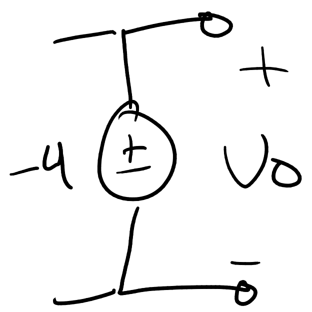
D_1 is off D_2 ON



Example

For simplicity, assume ideal diodes





Solution

For $t=0$, $V_s=0$ and both D_1 and D_2 are reverse biased

For $0<V_s<2$, D_1 and D_2 remain off, $V_o=V_s$

For $V_s>2$, D_1 turns ON and

$$-V_s+i_1(10k\Omega)+i_1(10k\Omega)+2=0$$

$$\Rightarrow i_1=\frac{V_s-2}{20k\Omega}$$

$$V_o=i_1R_2+2 \Rightarrow V_o=\frac{V_s-2}{20k\Omega}(10k\Omega)+2$$

$$\Rightarrow V_o=\frac{1}{2}(V_s-2)+2 \Rightarrow \frac{1}{2}V_s+1$$

If $V_s=6$, $V_o=4V$

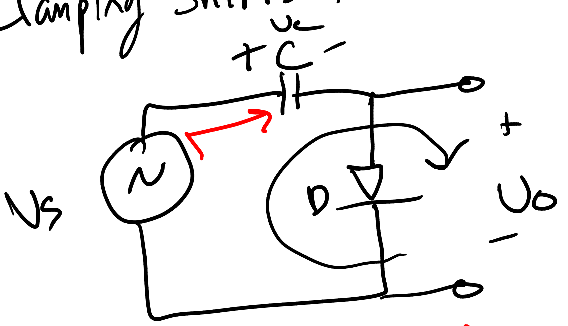
For $-4<V_s<0$, D_1 and D_2 are off and $V_o=V_s$

For $V_s\le -4$ D_2 turns ON

If D_1 is off and D_2 is ON, $V_o=-4V$

Clamper Circuits

* Clamping shifts the entire signal voltage by a DC level.



Assume the capacitor is initially uncharged and ideal diode

During the first positive half-cycle $V_c=V_s$

* once peak value is reached, Vs begins to decrease and the diode becomes reverse biased

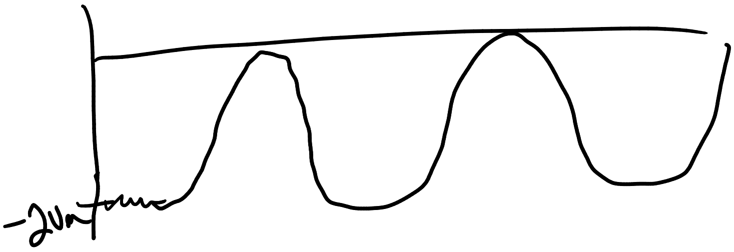
* Ideally, C can't discharge so $V_C = V_m$

* From KVL

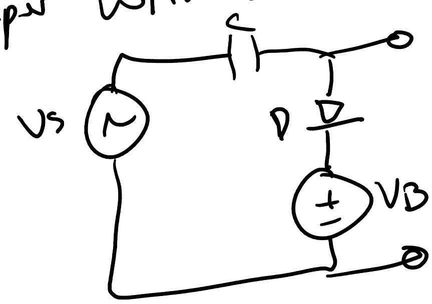
$$V_o = V_s - V_c = V_m \sin(\omega t) - \underline{V_m}$$

DC offset

* The output voltage is "clamped" at 0V or $V_o \le 0$



Clamper with a DC Source



For this circuit, the output is clamped at VB

Since $V_B = V_o$

End of Diodes

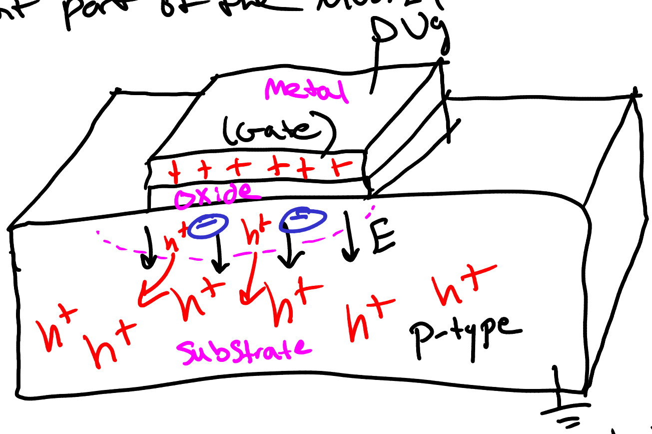
Fundamentals of MOSFETS

Metal Oxide Semiconductor Field effect transistor

MOS Capacitor

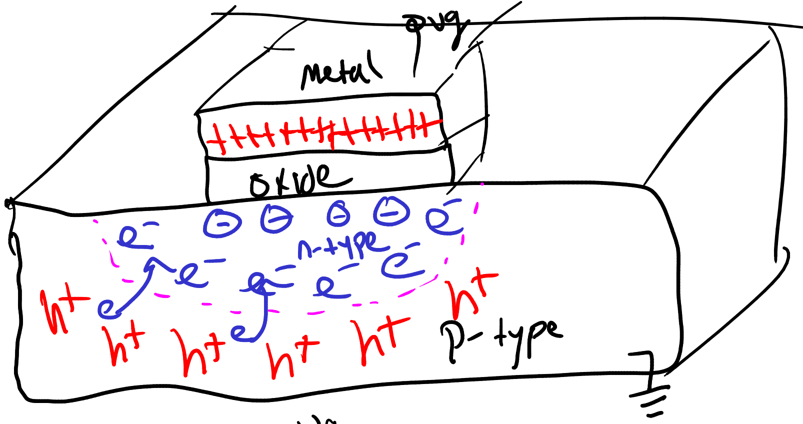
* The most important part of the MOSFET

* What happens when we apply a voltage



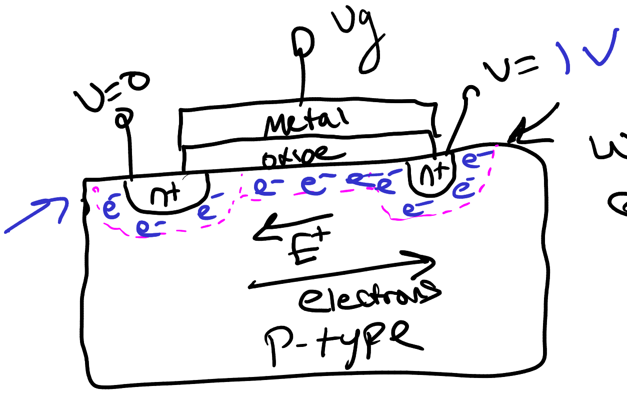
This electric field repels the positively charged holes, artifically making a depletion region. This is depletion mode

* What happens for larger Vg



This is called inversion

MOSFET Design



"source" of electrons

where the excess electron "drain"

MOSFET have 3 pins

- Gate
- Source
- Drain

This is an n-channel MOSFET

*Four Basic MOSFET Structures ← just looked at

* n-channel enhancement mode
* p-channel enhancement mode
* n-channel depletion mode
* p-channel depletion mode